SAT Resetmaxxing for Fun and Profit

(Final Title TBD)

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Abstract

We improve SATCoP performance by resetting the connection tableau often. This causes more and more varied SAT statements to be added to the SAT prover, which causes it to saturate faster.

1 Introduction

2 Preliminaries

(TODO: Tableau or Tableaux? Bibliography is inconsistent.)

2.1 Tableau Solving Methods

Minimalist tableau-style provers based on Prolog have been used with great success since the release of leanTAP in 1995[1], and have consistently achieved high rankings in solving FOF and CNF problems in the Thousands of Problems for Theorem Provers (TPTP) library[7].

Rather than working like saturation provers such as Vampire[5], which repeatedly derive new clauses from the current clause set until they derive a contradiction or reach saturation, tableau-based provers attempt to create a Connection Tableau with ground terms. The prover builds a tree, where each node corresponds to a particular clause of the problem in conjunctive normal form, using one of the following four steps.

(TODO: Add a diagram and/or a description.)

Figure 1: Description of the operations of Tableau calculus[2].

LeanCoP[4] improves
Talk about LeanCoP[3], possibly MeanCoP[2].

2.2 SATCoP and Similar Methods

3 Research Vehicle

Talk about the Connections Prover[6].

4 Experimental Evaluation

5 Future Results

References

[1] Bernhard Beckert and Joachim Posegga. leanTAP: Lean tableau-based deduction. 15(3):339–358, 1995.

[2] Michael Färber. A Curiously Effective Backtracking Strategy for Connection Tableaux, 2024.

[3] Jens Otten. Restricting backtracking in connection calculi. 23(2–3):159–182, 2010.

[4] Jens Otten and Wolfgang Bibel. leanCoP: Lean connection-based theorem proving. 36(1–2):139–161, 2003.

[5] Alexandre Riazanov and Andrei Voronkov. The design and implementation of VAMPIRE. 15(2–3):91–110, 2002.

[6] Fredrik Rømming, Jens Otten, and Sean B. Holden. Connections: Markov Decision Processes for Classical, Intuitionistic and Modal Connection Calculi. In *Proceedings of the 1st International Workshop on Automated Reasoning with Connection Calculi*, volume 3613 of *CEUR Workshop Proceedings*, pages 107–118, 2023.

[7] Geoff Sutcliffe. The 12th IJCAR automated theorem proving system competition—CASC-J12. 38(1):3–20, 2025.